

The Impact of Gamification for Science Technology Engineering Mathematics (STEM) in South Africa

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Abstract

Most gamified use-cases transpire in the global North, with proven benefits in customized learning. However, the adoption and implementation of gamification in LMICs has been slow, due to economic induced disparities. This study, thus, aims to integrate gamification in a STEM application developed for LMICs. The study analyses the global trends from 2004-2025 to represent the changes in . The primary objective is to assess the global trends in education curriculum maturity.

Key words: Gamification, Curriculum Digital Maturity

Introduction

Science, technology, engineering and mathematics in South Africa, outlined in the National development Plan. The South African government seeks to improve the performance in STEM subjects. The underperformance...

Literature review

The emergence of Gamification

Teaching has evolved immensely over time, and this evolution has been influenced by numerous philosophical contributions such as Paolo Freire (1968), who introduced the radical idea of critical pedagogy. This key idea justifying the thought suggested that suggests education needed to evolve from the traditional confines of the banking model. This necessitated evolution has come in other forms since. Gamification is one of such forms which has risen to global prominence, following the seminal contributions of Thomas Malone, who laid the foundation for game mechanics for motivation, and learning (Malone, 1981). The framework outlined the benefits of supervised learning long before gamification was popularised by the scholar Nick Pelling. The literature in this area highlighted how game mechanics, and conventional game design and its reward based learning approach incentivised learning and the subsequent performance. Some instances of this in practice is represented in the works by James Paul Gee who is famous for using the XP point system, typically seen in games, as the grading system in his class room.

Gamification in the African context

In Africa, the historical legacies and economic disparities have been factors which have potentially nullified the gains from this technology. Therefore, it is arguably within the interest of African Countries to help empower its population through the provision of access to state-sponsored gamified learning platforms.

Zooming in on Gamification in South Africa

In South Africa Gamification has been popularised, however, the contextual factors such as teaching capacity, infrastructural capacity, data cost has acted as a barrier of entry. Despite these challenges, studies which outline these gains have still risen to prominence. A study of such note is by Botha-Ravyse, Lennox and Jordaan (2018). These scholars developed an application to help assess how mobile applications could harness application development for gamification for sports and

recreational management. The application was an android application called Ama-Zing Race which rewarded and penalised wrong answer.

Their study found that from a sample size of $n=85$, that 61% of the pool identified technical issues, whilst 27% found the application as an inhibition to their studies, whilst 12% found the application as useful. These worrisome numbers from their study suggest that their application had development issues and as such is not a fair assessment of the true gains that can be harnessed from gamification.

Furthermore, their study outlined complaints by the students, the students critiqued their studies difficulty of questions arguing the difficulty failed to adequately prepare them for their studies. However, this account is contradicts the 'fruitful order' of question difficulty outlined by Gee (2003). Gee argued that the order of question mattered in gamification. His principal argument was that if the question difficulty is too high, it can lead to issues learning. He proposed a gradual approach to questions. These scholars followed this, however, the ease of questions did not gradually increase in the manner recommended by Gee.

Gamification is therefore an approach that requires a systemic framework that can help to create structured learning instead of haphazard delivery. These sentiments are raised in the paper by Botha-Ravyse, Lennox and Jordaan (2018), however, their approach to assessment design lacked such a structure.

The gap this study seeks to address partains to the neccessitated framework adoption that addresses these literary gaps and social gaps through the democratization of structured gamified learning.

Gamification Issues

Botha-Ravyse, Lennox and Jordaan (2018) outlined that gamification assessments require a framework to uphold assessment quality and standards. The lack of standardised criterion leads to challenges in research pertaining to aspects such as game-element definitions (Ofosu-Ampong, 2020). This compounded by the variance in games also posses a challenge when it comes to replicating the studies. Relative difficulty also posses a challenge because as showcased by Botha-Ravyse, Lennox and Jordaan, some assessments could lack the sufficient difficulty needed to stimulate learning.

Challenges in the South African context

Harnessing Gamification

The local context

Methodology

This study utilises a quantitative description design into the concept of 'Gamification'. This study utilised a google trends. The data was gathered using secondary data from google trends, through the consistent use of the same parameter. This parameter was used to geographically focus on the the globe, South Africa and the South African provinces. The period of this study ranges from January 2004 to October 2025. This chronological assessments is used to showcase the global interest in gamification over-time. This interest over-time is represented through a time-series analysis into the google-Trends data.

Research Aims:

The search interest is...

Research Questions:

1. *What has been the global interest in Gamification since 2004?*
2. *What has been the South African interest in Gamification since 2004?*
3. *What has been the South African Provincial interest in gamification?*

The data process design:

1. The study used the time range from 2004-2025.
2. The search string used was Gamification.
3. The study used the Pytrends API to source the date.
4. The output of chronological time-series visualisations was used to assess the global trends over-time.
5. The visualisation were generated using google sheets.

Findings

Figure 1, represents the global interest in gamification following its pronounced popularisation in 2002. However, the spike in interest is only showcased interest from the 2010s to the present. Figure 2 represents the zoomed in analysis into South Africa. The findings account for a rapid increase in interest in 2015. The general interest has been fluctuating constantly over-time. Figure 3. The local South African context the provincial interest has been largely concentrated in Gauteng and the Western Cape.

Figure 1. Global

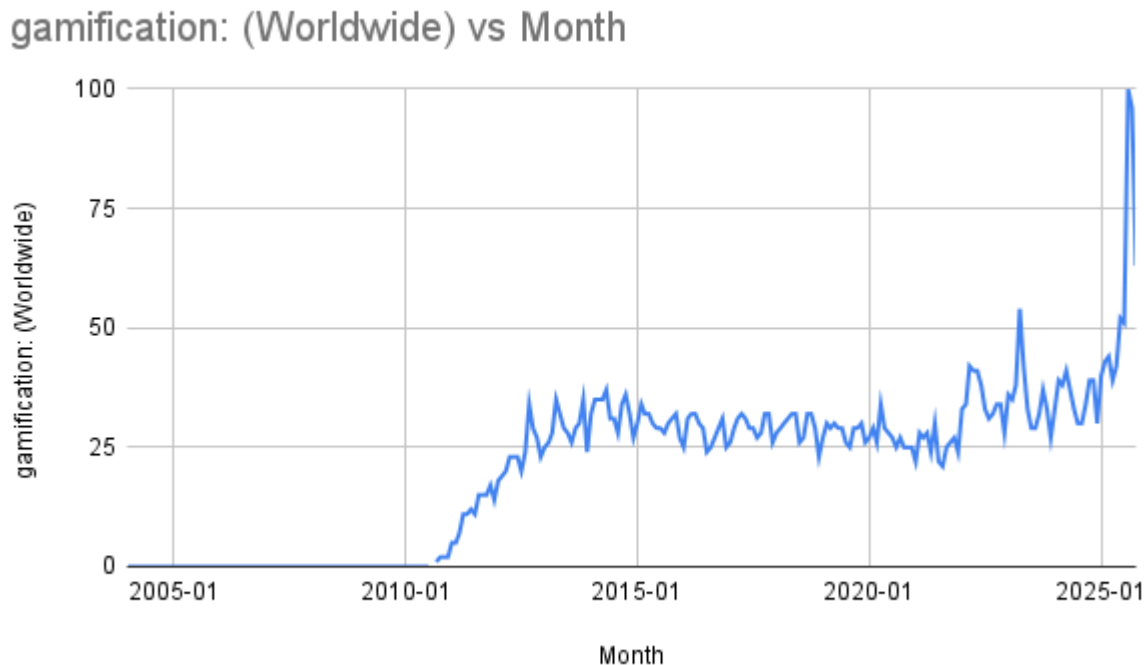
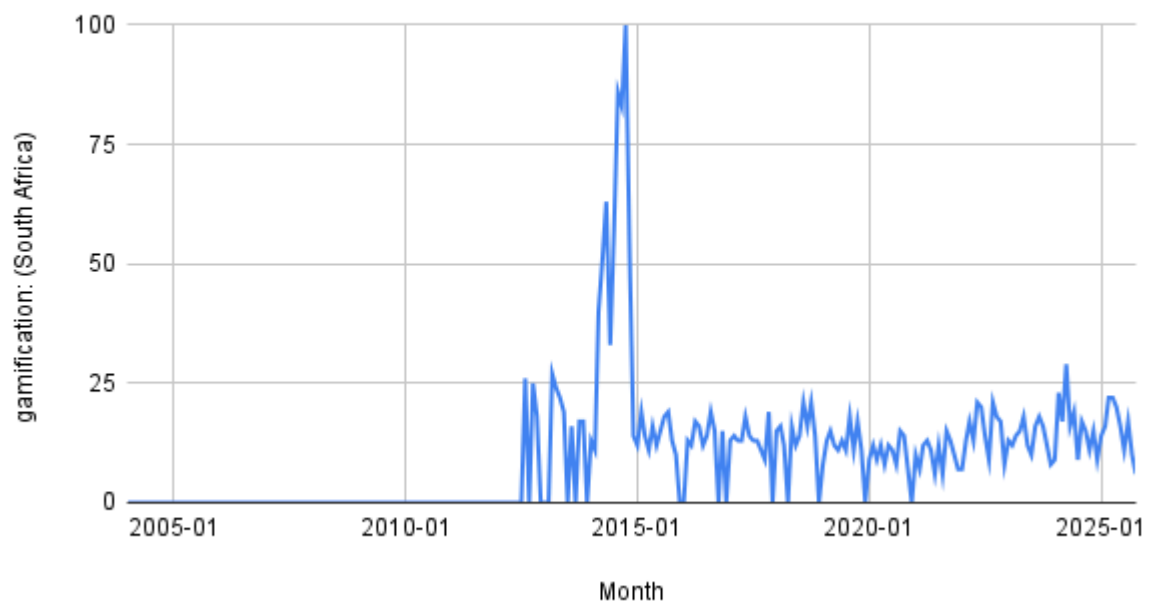


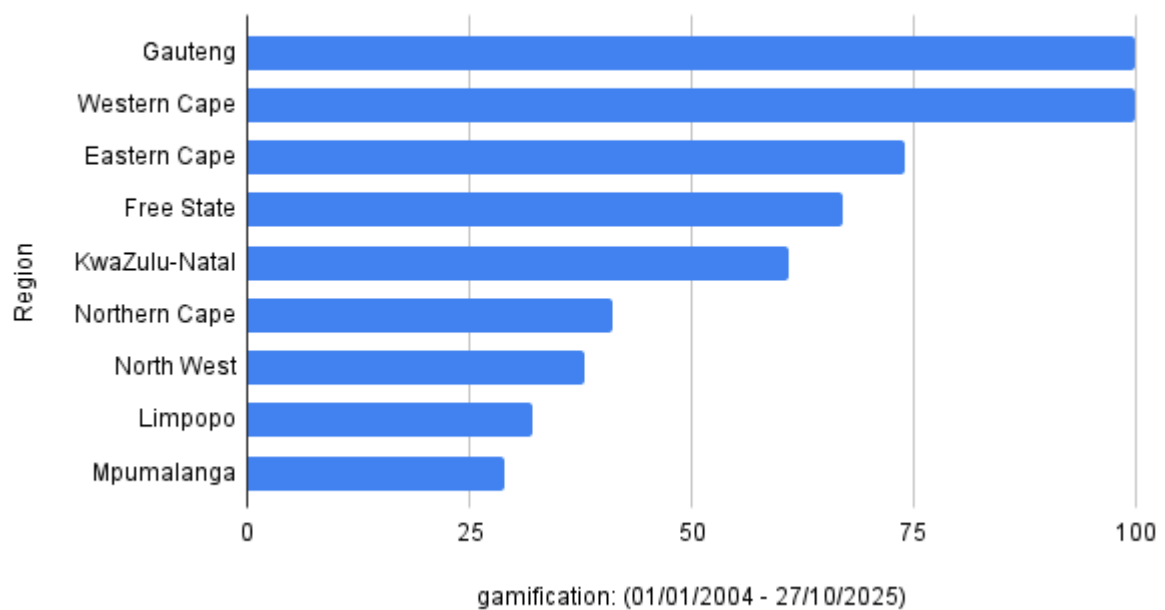
Figure 2. South Africa

gamification: (South Africa) vs Month



South African Provinces Interest in Gamification

gamification: (01/01/2004 - 27/10/2025) vs Region



Discussion

Limitations of the study

The study is geographically confined by the location. This constriction limits the analysis and further studies could explore other African countries.

Conclusion

Reference List

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Gee, James. (2003). What Video Games Have to Teach Us About Learning and Literacy. Computers in Entertainment. 1. 20. 10.1145/950566.950595.